

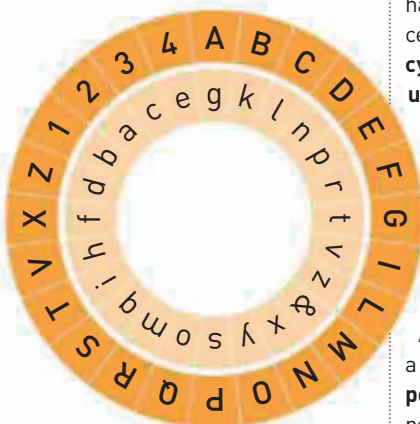


An illuminated page from Gutenberg's 42-line Bible. The 48 surviving copies are among the most valuable books in the world.

IN 1454, GERMAN PRINTER JOHANNES GUTENBERG

completed his edition of the Bible printed with 42 lines on each page. It was the first substantial book printed in Europe and its 180 or so copies sold out almost immediately. The Gutenberg Bible was soon followed by hundreds of works by Gutenberg and other printers, allowing the much **more rapid dissemination of scientific ideas**.

In 1464, German mathematician **Johannes Müller**, also known as Regiomontanus (1436–76), completed his *De Triangulis Omnimodis* (*On Triangles*), a systematic **textbook for trigonometry**. One of his fundamental propositions



Cracking the code

Alberti's disk operated by rotating the inner ring so that an agreed character (such as "g") lined up with the A of the outer ring.

30 FLORINS THE ORIGINAL COST OF THE GUTENBERG BIBLE

was that two triangles that have sides in similar proportions will also have similar angles. He relied on the work of Arabic mathematicians.

The first work on cryptography had been written in the 13th century. By the 15th century, **cyphers were in widespread use for diplomatic correspondence**. Codes relied on **monoalphabetic substitution**, in which each letter is transformed into the same encoded letter. In 1466, Italian painter and philosopher **Leon Battista Alberti** (1404–72) devised a cypher disk that made **polyalphabetic substitution** possible. Each new rotation of the disk generated an entirely new alphabet for coding.



1454 Johannes Gutenberg prints his 42-line Bible

1456 Italian astronomer Paolo Toscanelli observes what is later known as Halley's Comet

1460 German army surgeon Heinrich von Pilschke describes gunshot wounds

1458 German philosopher Nicholas of Cusa expresses early version of the Law of Continuity

1466 Leon Battista Alberti writes treatise describing a cypher disk

1464 Regiomontanus writes a systematic textbook for trigonometry

1472 Austrian astronomer Georg Peurbach publishes *New Theories of the Planets*—the first widely circulated printed astronomical work

1478 Publication of the *Trivium Arithmetic*, the first printed general work on mathematics